

Self-Supervised Pre-Training of Swin Transformers for 3D Medical Image Analysis

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Abstract

Vision Transformers (ViT)s have shown great performance in self-supervised learning of global and local representations that can be transferred to downstream applications. Inspired by these results, we introduce a novel self-supervised learning framework with tailored proxy tasks for medical image analysis. Specifically, we propose: (i) a new 3D transformer-based model, dubbed Swin UNet Transformers (Swin UNETR), with a hierarchical encoder for self-supervised pre-training; (ii) tailored proxy tasks for learning the underlying pattern of human anatomy. We demonstrate successful pre-training of the proposed model on 5,050 publicly available computed tomography (CT) images from various body organs. The effectiveness of our approach is validated by fine-tuning the pre-trained models on the Beyond the Cranial Vault (BTCV) Segmentation Challenge with 13 abdominal organs and segmentation tasks from the Medical Segmentation Decathlon (MSD) dataset. Our model is currently the state-of-the-art on the public test leaderboards of both MSD¹ and BTCV² datasets. Code: <https://monai.io/research/swin-unetr>.

1. Introduction

Vision Transformers (ViT)s [16] have started a revolutionary trend in computer vision [12, 54] and medical image analysis [6, 19]. Transformers demonstrate exceptional capability in learning pre-text tasks, are effective in learning of global and local information across layers, and provide

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¹<https://decathlon-10.grand-challenge.org/evaluation/challenge/leaderboard/>

²<https://www.synapse.org/#!/Synapse:syn3193805/wiki/217785/>

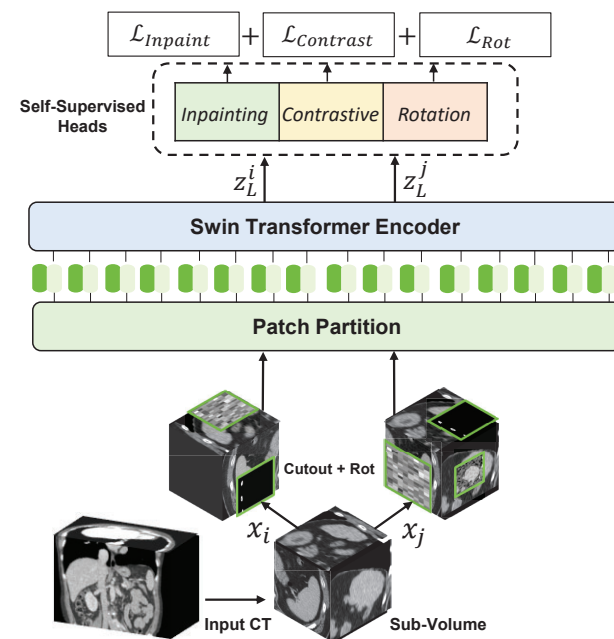


Figure 1. Overview of our proposed pre-training framework. Input CT images are randomly cropped into sub-volumes and augmented with random inner cutout and rotation, then fed to the Swin UNETR encoder as input. We use masked volume inpainting, contrastive learning and rotation prediction as proxy tasks for learning contextual representations of input images.

scalability for large-scale training [38, 52]. As opposed to convolutional neural networks (CNNs) with limited receptive fields, ViTs encode visual representations from a sequence of patches and leverage self-attention blocks for modeling long-range global information [38]. Recently, Shifted windows (Swin) Transformers [30] proposed a hierarchical ViT that allows for local computing of self-attention with

SwinTransformers 的自监督预训练用于三维医学 图像分析

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摘要

视觉Transformer (ViT) 在全局和局部表征的自监督学习中展现出卓越性能，这些表征可迁移至下游应用。受此启发，我们提出了一种新颖的自监督学习框架，并针对医学图像分析设计了定制化的代理任务。具体而言，我们提出：(i) 一种新的基于Transformer的三维模型，称为Swin UNet Transformers (Swin UNETR)，其配备分层编码器用于自监督预训练；(ii) 针对学习人体解剖结构内在模式的定制化代理任务。我们在来自多种身体器官的5,050张公开计算机断层扫描 (CT) 图像上成功实现了所提模型的预训练。通过在Beyond the Cranial Vault (BTCV) 分割挑战赛中的13个腹部器官以及医学分割十项全能 (MSD) 数据集的分割任务上微调预训练模型，验证了方法的有效性。我们的模型目前在MSD¹和BTCV²数据集的公开测试排行榜上均处于最先进水平。代码：<https://monai.io/research/swin-unetr>。

1. 引言

视觉Transformer (ViT) [16] 在计算机视觉领域开启了一场革命性的浪潮 [12, 54] 以及医学图像分析 [6, 19]。Transformer在学习前置任务方面展现出卓越的能力，能够有效学习跨层的全局和局部信息，并提供

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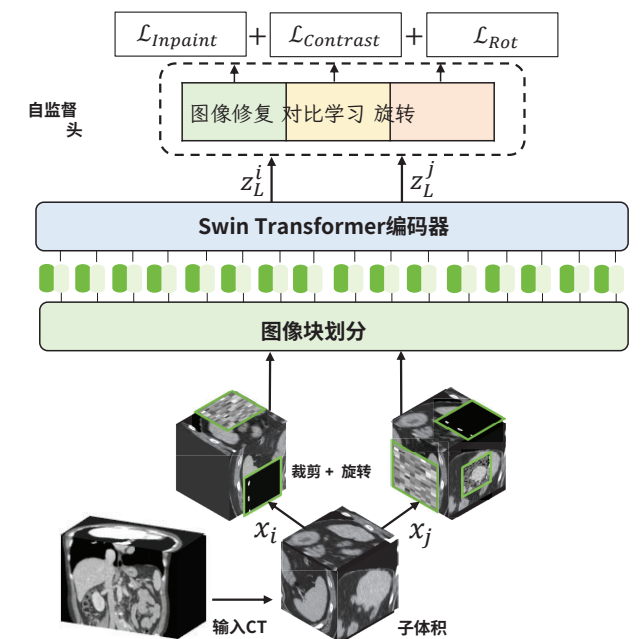


图1. 我们提出的预训练框架概述。输入CT图像被随机裁剪为子体积，并通过随机内部裁剪和旋转进行增强，然后作为输入馈送到Swin UNETR编码器。我们使用掩码体积修复、对比学习和旋转预测作为代理任务，以学习输入图像的上下文表示。

大规模训练的可扩展性 [38, 52]。与感受野有限的卷积神经网络 (CNNs) 不同，ViT从一系列图像块中编码视觉表示，并利用自注意力模块对长距离全局信息进行建模 [38]。最近，移位窗口 (Swin) Transformer [30] 提出了一种分层视觉Transformer，允许通过以下方式进行局部自注意力计算：

non-overlapping windows. This architecture achieves linear complexity as opposed to quadratic complexity of self-attention layers in ViT, hence making it more efficient. In addition, due to the hierarchical nature of Swin Transformers, they are well-suited for tasks requiring multi-scale modeling.

In comparison to CNN-based counterparts, transformer-based models learn stronger features representations during pre-training, and as a result perform favorably on fine-tuning downstream tasks [38]. Several recent efforts on ViTs [5, 48] have achieved new state-of-the-art results by self-supervised pre-training on large-scale datasets such as ImageNet [15].

In addition, medical image analysis has not benefited from these advances in general computer vision due to: (1) large domain gap between natural images and medical imaging modalities, like computed tomography (CT) and magnetic resonance imaging (MRI); (2) lack of cross-plane contextual information when applied to volumetric (3D) images (such as CT or MRI). The latter is a limitation of 2D transformer models for various medical imaging tasks such as segmentation. Prior studies have demonstrated the effectiveness of supervised pre-training in medical imaging for different applications [9, 39]. But creating expert-annotated 3D medical datasets at scale is a non-trivial and time-consuming effort.

To tackle these limitations, we propose a novel self-supervised learning framework for 3D medical image analysis. First, we propose a new architecture dubbed Swin UNet Transformers (Swin UNETR) with a Swin Transformer encoder that directly utilizes 3D input patches. Subsequently, the transformer encoder is pre-trained with tailored, self-supervised tasks by leveraging various proxy tasks such as image inpainting, 3D rotation prediction, and contrastive learning (See Fig. 1 for an overview). Specifically, the human body presents naturally consistent contextual information in radiographic images such as CT due to its depicted anatomical structure [43, 50]. Hence, proxy tasks are utilized for learning the underlying patterns of the human anatomy. For this purpose, we extracted numerous patch queries from different body compositions such as head, neck, lung, abdomen, and pelvis to learn robust feature representations from various anatomical contexts, organs, tissues, and shapes.

Our framework utilizes contrastive learning [35], masked volume inpainting [37], and 3D rotation prediction [17] as pre-training proxy tasks. The contrastive learning is used to differentiate various ROIs of different body compositions, whereas the inpainting allows for learning the texture, structure and correspondence of masked regions to their surrounding context. The rotation task serves as a mechanism to learn the structural content of images and generates various sub-volumes that can be used for contrastive learning. We utilize these proxy tasks to pre-train our proposed framework on a collection of 5,050 CT images that are acquired from various publicly available datasets.

Furthermore, to validate the effectiveness of pre-training,

we use 3D medical image segmentation as a downstream application and reformulate it as a 1D sequence-to-sequence prediction task. For this purpose, we leverage the Swin UNETR encoder with hierarchical feature encoding and shifted windows to extract feature representations at four different resolutions. The extracted representations are then connected to a CNN-based decoder. A segmentation head is attached at the end of the decoder for computing the final segmentation output. We fine-tune Swin UNETR with pre-trained weights on two publicly available benchmarks of Medical Segmentation Decathlon (MSD) and the Beyond the Cranial Vault (BTCV). Our model is currently the state-of-the-art on their respective public test leaderboards.

Our main contributions in this work are summarized as follows:

- We introduce a novel self-supervised learning framework with tailored proxy tasks for pre-training on CT image datasets. To this end, we propose a novel 3D transformer-based architecture, dubbed as Swin UNETR, consisting of an encoder that extracts feature representations at multiple resolutions and is utilized for pre-training.
- We demonstrate successful pre-training on a cohort of 5,050 publicly available CT images from various applications using the proposed encoder and proxy tasks. This results in a powerful pre-trained model with robust feature representation that could be utilized for various medical image analysis downstream tasks.
- We validate the effectiveness of proposed framework by fine-tuning the pre-trained Swin UNETR on two public benchmarks of MSD and BTCV and achieve *state-of-the-art* on the test leaderboards of both datasets.

2. Related Works

Medical Segmentation with Transformers Vision transformers are first used in classification tasks and are adopted from sequence-to-sequence modeling in natural language processing. Self-attention mechanisms that aggregate information from the entire input sequence are first achieving comparable, then better performance against prior arts of convolutional architectures such as ResNet [21] or U-Net [13]. Recently, transformer-based networks [24, 47, 49, 55] are proposed for medical image segmentation. In these pioneering works, the transformer blocks are used either as a bottleneck feature encoder or as additional modules after convolutional layers, resulting in limited exploitation of the spatial context advantages of transformers. Comparing to prior works [6, 47], which are using transformers as secondary encoder, we propose to utilize transformers to embed high-dimensional volumetric medical images, which allow for a more direct encoding of 3D patches and positional embeddings.

非重叠窗口。该架构实现了线性复杂度，而ViT中自注意力层的复杂度为二次复杂度，因此使其更加高效。此外，由于Swin Transformer的分层特性，它们非常适合需要多尺度建模的任务。

与基于CNN的对应模型相比，基于Transformer的模型在预训练期间学习到更强的特征表示，因此在微调下游任务时表现更优 [38]。近期关于视觉Transformer的多项研究 [5, 48]通过在ImageNet等大规模数据集上进行自监督预训练，取得了新的最先进水平结果 [15]。

此外，医学影像分析未能从计算机视觉的这些进展中受益，原因在于：(1) 自然图像与医学成像模式（如计算机断层扫描（CT）和磁共振成像（MRI））之间存在巨大的域差距；(2) 应用于体积（3D）图像（如CT或MRI）时缺乏跨平面上下文信息。后者是2D Transformer模型在各种医学成像任务（如分割）中的局限性。先前的研究已证明监督预训练在医学影像中对不同应用的有效性 [9, 39]。但大规模创建专家标注的3D医学数据集是一项非平凡且耗时的努力。

为了解决这些局限性，我们提出了一种新颖的自监督学习框架，用于三维医学图像分析。首先，我们提出了一种名为Swin UNet Transformers（Swin UNETR）的新架构，其配备的Swin Transformer编码器可直接利用3D输入图像块。随后，该Transformer编码器通过利用图像修复、3D旋转预测和对比学习等多种代理任务，以定制

我们的框架利用对比学习 [35]，掩码体积修复 [37]，和3D旋转预测 [17] 作为预训练代理任务。对比学习用于区分不同身体部位的各种感兴趣区域，而图像修复则允许学习掩码区域的纹理、结构及其与周围上下文的对应关系。旋转任务作为一种学习图像结构内容的机制，并生成可用于对比学习的各种子体积。我们利用这些代理任务，在从各种公开数据集获取的5,050张CT图像集合上预训练我们提出的框架。

此外，为了验证预训练的有效性，

我们将3D医学图像分割作为下游应用，并将其重新表述为1D序列到序列预测任务。为此，我们利用具有分层特征编码和移位窗口的Swin UNETR编码器，在四种不同分辨率下提取特征表示。然后，提取的表示被连接到基于CNN的解码器。解码器末端附加了一个分割头，用于计算最终的分割输出。我们在医学分割十项全能（MSD）和Beyond the Cranial Vault（BTCV）两个公开基准上，使用预训练权重对Swin UNETR进行微调。我们的模型目前在各自的公开测试排行榜上处于最先进水平。

我们在这项工作中的主要贡献总结如下：

- 我们提出了一种新颖的自监督学习框架，并针对CT图像数据集预训练设计了定制的代理任务。为此，我们提出了一种新颖的基于3D Transformer的架构，称为Swin UNETR，其包含一个编码器，能够在多个分辨率上提取特征表示，并用于预训练。
- 我们展示了利用所提出的编码器和代理任务，在来自不同应用的5,050张公开可用CT图像队列上成功进行了预训练。这产生了一个强大的预训练模型，具有稳健的特征表示，可用于各种医学图像分析下游任务。
- 我们通过在MSD和BTCV两个公开基准上微调预训练的Swin UNETR来验证所提出框架的有效性，并在两个数据集的测试排行榜上达到了最先进水平。

2. 相关工作

基于Transformer的医学分割视觉Transformer最初用于分类任务，源自自然语言处理中的序列到序列建模。从整个输入序列中聚合信息的自注意力机制起初达到了与ResNet [21]或U-Net [13]等卷积架构先前工作相当的性能，随后则实现了更优的表现。近来，基于Transformer的网络 [24, 47, 49, 55]被提出用于医学图像分割。在这些开创性工作中，Transformer模块要么被用作瓶颈特征编码器，要么被用作卷积层之后的附加模块，导致Transformer在空间上下文方面的优势未能得到充分利用。与将Transformer用作次级编码器的先前工作 [6, 47]相比，我们提出利用Transformer来嵌入高维体积医学图像，从而能够更直接地编码3D图像块和位置嵌入。

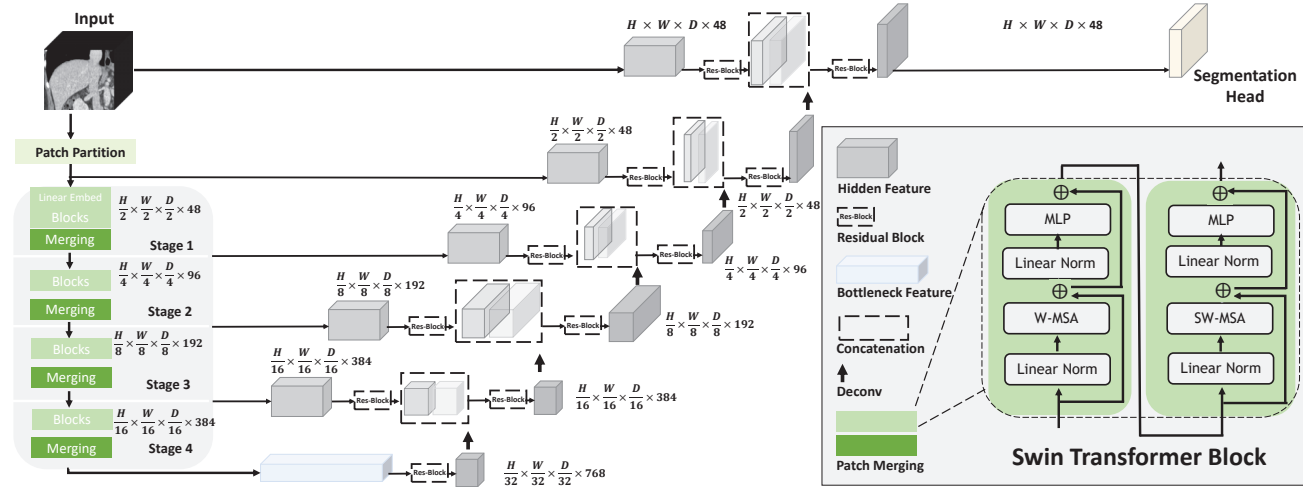


Figure 2. Overview of the Swin UNETR architecture.

Most medical image analysis tasks such as segmentation requires dense inference from multi-scale features. Skip connection-based architectures such as UNet [13] and pyramid networks [40] are widely adopted to leverage hierarchical features. However, vision transformers with a single patch size, while successful in natural image applications, are intractable for high-resolution and high-dimensional volumetric images. To avoid quadratic overflow of computing self-attention at scales [29, 41], Swin Transformer [30, 31] is proposed to construct hierarchical encoding by a shifted-window mechanism. Recent works such as Swin UNet [4] and DS-TransUNet [28] utilize the merits of Swin Transformers for 2D segmentation and achieve promising performance. Augmenting the above-mentioned methods, we learn from 3D anatomy in broader medical image segmentation scenarios by incorporating hierarchically volumetric context.

Pre-training in Medical Image Analysis In medical image analysis, previous studies of pre-training on labeled data demonstrate improved performance by transfer learning [9, 39]. However, generating annotation for medical images is expensive and time-consuming. Recent advances in self-supervised learning offer the promise of utilizing unlabeled data. Self-supervised representation learning [2, 14, 27] constructs feature embedding spaces by designing pre-text tasks, such as solving jigsaw puzzles [34]. Another commonly used pre-text task is to memorize spatial context from medical images, which is motivated by image restoration. This idea is generalized to inpainting tasks [18, 37, 57] to learn visual representations [3, 7, 46] by predicting the original image patches. Similar efforts for reconstructing spatial context have been formulated as solving Rubik’s cube problem [58], random rotation prediction [17, 42] and contrastive coding [11, 35]. Different from these efforts, our pre-training framework is simulta-

neously trained with a combination of pre-text tasks, tailored for 3D medical imaging data, and leverages a transformer-based encoder as a powerful feature extractor.

3. Swin UNETR

Swin UNETR comprises a Swin Transformer [30] encoder that directly utilizes 3D patches and is connected to a CNN-based decoder via skip connections at different resolutions. Fig. 2 illustrates the overall architecture of Swin UNETR. We describe the details of encoder and decoder in this section.

3.1. Swin Transformer Encoder

Assuming that the input to the encoder is a sub-volume $\mathcal{X} \in \mathbb{R}^{H \times W \times D \times S}$, a 3D token with a patch resolution of (H', W', D') has a dimension of $H' \times W' \times D' \times S$. The patch partitioning layer creates a sequence of 3D tokens with size $\frac{H}{H'} \times \frac{W}{W'} \times \frac{D}{D'}$ that are projected into a C -dimensional space via an embedding layer. Following [30], for efficient modeling of token interactions, we partition the input volumes into non-overlapping windows and compute local self-attention within each region. Specifically, at layer l , we use a window of size $M \times M \times M$ to evenly divide a 3D token into $\left\lceil \frac{H'}{M} \right\rceil \times \left\lceil \frac{W'}{M} \right\rceil \times \left\lceil \frac{D'}{M} \right\rceil$ windows. In the subsequent layer $l+1$, we shift the partitioned windows by $(\lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor)$ voxels. The shifted windowing mechanism is illustrated in Fig. 3. The outputs of encoder blocks in layers l and $l+1$ are computed as in

$$\begin{aligned} \hat{z}^l &= \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1} \\ \hat{z}^l &= \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l \\ \hat{z}^{l+1} &= \text{SW-MSA}(\text{LN}(z^l)) + z^l \\ \hat{z}^{l+1} &= \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}, \end{aligned} \quad (1)$$

where W-MSA and SW-MSA denote regular and window partitioning multi-head self-attention modules, respectively,

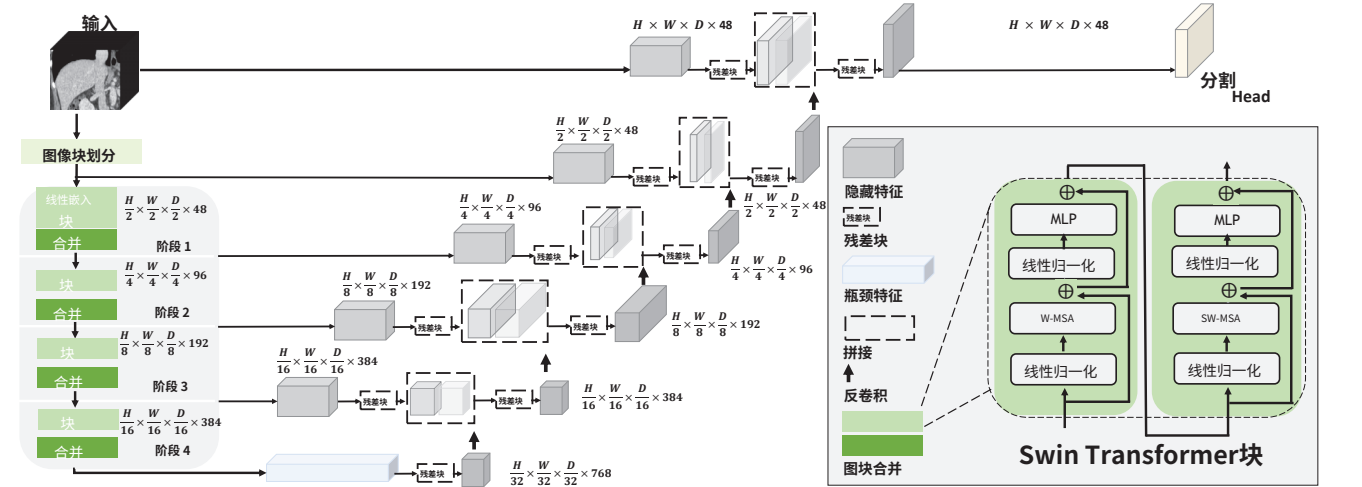


图 2. Swin UNETR 架构概览

e.

大多数医学图像分析任务（如分割）需要从多尺度特征中进行密集推理。基于跳跃连接的架构（如 UNet [13] 和金字塔网络 [40]）被广泛采用以利用层次化特征。然而，采用单一图像块大小的视觉Transformer虽然在自然图像应用中取得了成功，但难以处理高分辨率和高维体积图像。为避免自注意力计算在尺度上的二次爆炸 [29, 41], Swin Transformer [30, 31]被提出通过移位窗口机制构建层次化编码。近期工作如Swin UNet [4]和DS-TransUNet [28]利用Swin Transformer的优势进行2D分割并取得了令人瞩目的性能。在上述方法的基础上，我们通过融入层次化体积上下文，在更广泛的医学图像分割场景中学习3D解剖结构。

医学图像分析中的预训练 在医学图像分析中，先前基于标注数据的预训练研究表明，通过迁移学习 [9, 39]可以提升性能。然而，为医学图像生成标注既昂贵又耗时。自监督学习的最新进展为利用无标注数据带来了希望。自监督表示学习 [2, 14, 27]通过设计前置任务来构建特征嵌入空间，例如解决拼图游戏 [34]。另一种常用的前置任务是记忆医学图像中的空间上下文，其灵感来源于图像修复。这一思想被推广到图像补全任务 [18, 37, 57]中，通过预测原始图像块来学习视觉表征 [3, 7, 46]。类似的用于重建空间上下文的工作已被表述为求解魔方问题 [58]、随机旋转预测 [17, 42]和对比编码 [11, 35]。与这些工作不同，我们的预训练框架同时

结合多种前置任务进行训练，专为3D医学影像数据量身定制，并利用基于Transformer的编码器作为强大的特征提取器。

3. Swin UNETR

SwinUNETR包含一个Swin Transformer [30]编码器，该编码器直接利用3D图像块，并通过不同分辨率下的跳跃连接与基于CNN的解码器相连。图2展示了Swin UNETR的整体架构。本节中我们将详细描述编码器和解码器的细节。

3.1. Swin Transformer编码器

假设编码器的输入是一个子体积 $\mathcal{X} \in \mathbb{R}^{H \times W \times D \times S}$ ，则图像块分辨率为 (H', W', D') 的3D标记具有 $H' \times W' \times D' \times S$ 的维度。图像块划分创建大小为 $\frac{H}{H'} \times \frac{W}{W'} \times \frac{D}{D'}$ 的3D标记序列，并通过嵌入层将其投影到 C 维空间中。遵循 [30], 以高效建模标记交互，我们将输入体数据划分为非重叠窗口，并在每个区域内计算局部自注意力。具体而言，在第 l 层，我们使用大小为 $M \times M \times M$ 的窗口将3D标记均匀划分为 $\left\lceil \frac{H'}{M} \right\rceil \times \left\lceil \frac{W'}{M} \right\rceil \times \left\lceil \frac{D'}{M} \right\rceil$ 个窗口。在后续的第 $l+1$ 层，我们将划分的窗口平移 $(\lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor)$ 个体素。移位窗口机制如图3所示。第 l 层和第 $l+1$ 层中编码器块的输出按照如下方式计算 —

$$\begin{aligned} \hat{z}^l &= \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1} \\ \hat{z}^l &= \text{MLP}(\text{LN}(\hat{z}^l)) + \hat{z}^l \\ \hat{z}^{l+1} &= \text{SW-MSA}(\text{LN}(z^l)) + z^l \\ \hat{z}^{l+1} &= \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}, \end{aligned} \quad (1)$$

其中 W-MSA 和 SW-MSA 分别表示常规多头自注意力模块和窗口划分多头自注意力模块，

$\hat{z}^l$ and $\hat{z}^l$ are the outputs of W-MSA and SW-MSA; LN and MLP denote layer normalization and Multi-Layer Perceptron (see Fig. 2). Following [30], we adopt a 3D cyclic-shifting for efficient batch computation of shifted windowing. Furthermore, we calculate the self-attention according to

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V. \quad (2)$$

where Q, K, V represent queries, keys and values respectively, d is the size of the query and key.

Our encoder uses a patch size of $2 \times 2 \times 2$ with a feature dimension of $2 \times 2 \times 2 \times 1 = 8$ (*i.e.* single input channel CT images) and a $C = 48$ -dimensional embedding space. Furthermore, the overall architecture of the encoder consists of 4 stages comprising of 2 transformer blocks at each stage (*i.e.* $L = 8$ total layers). In between every stage, a patch merging layer is used to reduce the resolution by a factor of 2. Stage 1 consists of a linear embedding layer and transformer blocks that maintain the number of tokens as $\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$. Furthermore, a patch merging layer groups patches with resolution $2 \times 2 \times 2$ and concatenates them, resulting in a $4C$ -dimensional feature embedding. A linear layer is then used to downsample the resolution by reducing the dimension to $2C$. The same procedure continues in stage 2, stage 3 and stage 4 with resolutions of $\frac{H}{4} \times \frac{W}{4} \times \frac{D}{4}$, $\frac{H}{8} \times \frac{W}{8} \times \frac{D}{8}$ and $\frac{H}{16} \times \frac{W}{16} \times \frac{D}{16}$ respectively. The hierarchical representations of the encoder at different stages are used in downstream applications such as segmentation for multi-scale feature extraction.

3.2. Decoder

The encoder of Swin UNETR is connected to a CNN-based decoder at each resolution via skip connections to create a “U-shaped” network for downstream applications such as segmentation. Specifically, we extract the output sequence representations of each stage i ($i \in \{0, 1, 2, 3, 4\}$) in the encoder as well as the bottleneck ($i = 5$) and reshape them into features with size $\frac{H}{2^i} \times \frac{W}{2^i} \times \frac{D}{2^i}$. The extracted representations at each stage are then fed into a residual block consisting of two post-normalized $3 \times 3 \times 3$ convolutional layers with instance normalization [45]. The processed features from each stage are then upsampled by using a deconvolutional layer and concatenated with processed features of the preceding stage. The concatenated features are fed into a residual block with aforementioned descriptions. For segmentation, we concatenate the output of the encoder (*i.e.* Swin Transformer) with processed features of the input volume and feed them into a residual block followed by a final $1 \times 1 \times 1$ convolutional layer with a proper activation function (*i.e.* softmax) for computing the segmentation probabilities (see Fig. 2 for details of the architecture).

4. Pre-training

We pre-train the Swin UNETR encoder with multiple proxy tasks and formulate it with a multi-objective loss func-

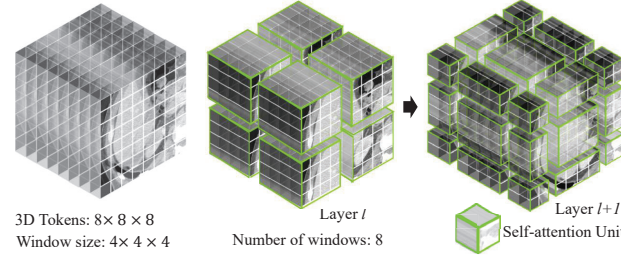


Figure 3. Shifted windowing mechanism for efficient self-attention computation of 3D tokens with $8 \times 8 \times 8$ tokens and $4 \times 4 \times 4$ window size.

tion (Fig. 1). The objective of self-supervised representation learning is to encode region of interests (ROI)-aware information of the human body. Inspired by previous works on context reconstruction [18, 57] and contrastive encoding [20], we exploit three proxy tasks for medical image representation learning. Three additional projection heads are attached to the encoder during pre-training. Furthermore, the downstream task, e.g. segmentation, fine-tunes the full Swin UNETR model with the projection heads removed. In training, sub-volumes are cropped random regions of the volumetric data. Then, stochastic data augmentations with random rotation and cutout are applied twice to each sub-volume within a mini-batch, resulting in two views of each data.

4.1. Masked Volume Inpainting

The cutout augmentation masks out ROIs in the sub-volume $\mathcal{X} \in \mathbb{R}^{H \times W \times D \times C}$ randomly with volume ratio of s . We attach a transpose convolution layer to the encoder as the reconstruction head and denote its output as $\hat{\mathcal{X}}^{\mathcal{M}}$. The reconstruction objective is defined by an $L1$ loss between $\mathcal{X}$ and $\hat{\mathcal{X}}^{\mathcal{M}}$

$$\mathcal{L}_{inpaint} = \|\mathcal{X} - \hat{\mathcal{X}}^{\mathcal{M}}\|_1, \quad (3)$$

The masked volume inpainting is motivated by prior work which focused on 2D images [37]. We extend it to 3D domain to showcase its effectiveness on representation learning of volumetric medical images.

4.2. Image Rotation

The rotation prediction task predicts the angle categories by which the input sub-volume is rotated. For simplicity, we employ R classes of 0° , 90° , 180° , 270° rotations along the z -axis. An MLP classification head is used for predicting the softmax probabilities $\hat{y}_r$ of rotation categories. Given the ground truth y_r , a cross-entropy loss is used for rotation prediction task:

$$\mathcal{L}_{rot} = -\sum_{r=1}^R y_r \log(\hat{y}_r), \quad (4)$$

$\hat{z}^l$ and $\hat{z}^l$ 分别是 W-MSA 和 SW-MSA 的输出；LN 和 MLP 分别表示层归一化和多层感知机（见图 2）。遵循 [30], 我们采用三维循环移位来实现移位窗口的高效批次计算。此外，我们根据以下方式计算自注意力

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V. \quad (2)$$

其中 Q, K, V 分别表示查询、键和值， d 是查询和键的大小。

我们的编码器使用的图像块大小为 $2 \times 2 \times 2$ ，特征维度为 $2 \times 2 \times 2 \times 1 = 8$ （即单输入通道 CT 图像），并使用 $C = 48$ 维嵌入空间。此外，编码器的整体架构由 4 个阶段组成，每个阶段包含 2 个 Transformer 模块（即 $L = 8$ 层）。在每个阶段之间，使用一个图像块合并层将分辨率降低 2 倍。阶段 1 由一个线性嵌入层和 Transformer 模块组成，这些模块维持标记数量为 $\frac{H}{2} \times \frac{W}{2} \times \frac{D}{2}$ 。此外，一个图像块合并层将分辨率为 $2 \times 2 \times 2$ 的图像块分组并拼接，得到一个 $4C$ 维特征嵌入。随后使用一个线性层将维度降至 $2C$ ，从而对分辨率进行下采样。相同的过程在阶段 2、阶段 3 和阶段 4 中继续进行，其分辨率分别为 $\frac{H}{4} \times \frac{W}{4} \times \frac{D}{4}$ ， $\frac{H}{8} \times \frac{W}{8} \times \frac{D}{8}$ 和 $\frac{H}{16} \times \frac{W}{16} \times \frac{D}{16}$ 。编码器在不同阶段的层次化表示被用于下游应用，例如分割中的多尺度特征提取。

3.2. 解码器

Swin UNETR 的编码器通过跳跃连接在每个分辨率上与基于 CNN 的解码器相连，形成一个用于下游应用（如分割）的“U 形”网络。具体而言，我们提取每个阶段的输出序列表示 i ($i \in \{0, 1, 2, 3, 4\}$)，在编码器以及瓶颈层 ($i = 5$) 中提取，并将它们重塑为大小为 $\frac{H}{2^i} \times \frac{W}{2^i} \times \frac{D}{2^i}$ 的特征。然后，每个阶段提取的表示被送入一个由两个后归一化的 $3 \times 3 \times 3$ 卷积层组成的残差块，并带有实例归一化 [45]。每个阶段处理后的特征随后通过反卷积层进行上采样，并与前一阶段处理后的特征进行拼接。拼接后的特征被送入具有上述描述的残差块。对于分割，我们将编码器的输出（即 Swin Transformer）与输入体数据处理后的特征进行拼接，并将它们送入一个残差块，随后是一个带有适当激活函数（即 softmax）的最终 $1 \times 1 \times 1$ 卷积层，用于计算分割概率（架构详情见图 2）。

4. 预训练

我们使用多个代理任务对 Swin UNETR 编码器进行预训练，并用多目标损失函数来构建

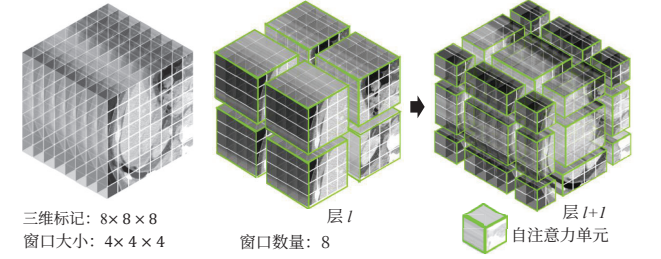


图 3. 用于 $8 \times 8 \times 8$ 个标记和 $4 \times 4 \times 4$ 窗口大小的三维标记高效自注意力计算的移位窗口机制。

该函数（图 1）。自监督表示学习的目标是对人体感兴趣区域（ROI）感知信息进行编码。受先前关于上下文重建 [18, 57] 和对比编码 [20] 工作的启发，我们利用三个代理任务进行医学图像表示学习。在预训练期间，编码器上附加了三个额外的投影头。此外，下游任务（如分割）在移除投影头的情况下对完整的 Swin UNETR 模型进行微调。在训练中，子体积是从体数据中随机裁剪的区域。然后，对小批量中的每个子体积应用两次带有随机旋转和裁剪的随机数据增强，从而得到每个数据的两个视图。

4.1. 掩码体积修复

裁剪增强以体积比 s 随机掩蔽子体积 $\mathcal{X} \in \mathbb{R}^{H \times W \times D \times C}$ 中的感兴趣区域。我们将一个转置卷积层附加到编码器作为重建头，并将其输出记为 $\hat{\mathcal{X}}^{\mathcal{M}}$ 。重建目标由 $\mathcal{X}$ 和 $\hat{\mathcal{X}}^{\mathcal{M}}$ 之间的 $L1$ 损失定义

$$\mathcal{L}_{inpaint} = \|\mathcal{X} - \hat{\mathcal{X}}^{\mathcal{M}}\|_1, \quad (3)$$

掩码体积修复受到先前工作的启发，这些工作聚焦于二维图像 [37]。我们将其扩展到三维领域，以展示其在体医学图像表征学习中的有效性。

4.2. 图像旋转

旋转预测任务通过输入子体积被旋转的角度类别来进行预测。为简单起见，我们采用 R 个类别，即沿 z 轴进行 0° 、 90° 、 180° 、 270° 旋转。使用 MLP 分类头来预测旋转类别的 softmax 概率 $\hat{y}_r$ 。给定真实标签 y_r ，使用交叉熵损失进行旋转预测任务：

$$\mathcal{L}_{rot} = -\sum_{r=1}^R y_r \log(\hat{y}_r), \quad (4)$$

The 3D rotation and cutout also serves simultaneously as an augmentation transformation for contrastive learning.

4.3. Contrastive Coding

The self-supervised contrastive coding presents promising performance on visual representation learning when transferred to downstream tasks [10, 36]. Given a batch of augmented sub-volumes, the contrastive coding allows for a better representation learning by maximizing the mutual information between positive pairs (augmented samples from same sub-volume), while minimizing that between negative pairs (views from different sub-volumes). The contrastive coding is obtained by attaching a linear layer to the Swin UNETR encoder, which maps each augmented sub-volume to a latent representation v . We use cosine similarity as the distance measurement of the encoded representations as defined in [10]. Formally, the 3D contrastive coding loss between a pair v_i and v_j is defined as:

$$\mathcal{L}_{contrast} = -\log \frac{\exp(\text{sim}(v_i, v_j)/t)}{\sum_k^{2N} 1_{k \neq i} \exp(\text{sim}(v_i, v_k)/t)}, \quad (5)$$

where t is the measurement of normalized temperature scale. 1 is the indicator function evaluating to 1 iff $k \neq i$. sim denotes the dot product between normalized embeddings. The contrastive learning loss function strengthens the intra-class compactness as well as the inter-class separability.

4.4. Loss Function

Formally, we minimize the total loss function by training Swin UNETR’s encoder with multiple pre-training objectives of masked volume inpainting, 3D image rotation & contrastive coding as follows:

$$\mathcal{L}_{tot} = \lambda_1 \mathcal{L}_{inpaint} + \lambda_2 \mathcal{L}_{contrast} + \lambda_3 \mathcal{L}_{rot}. \quad (6)$$

A grid-search hyper-parameter optimization was performed which estimated the optimal values of $\lambda_1 = \lambda_2 = \lambda_3 = 1$.

5. Experiments

5.1. Datasets

Pre-training Datasets : A total of 5 public CT datasets, consisting of 5,050 subjects, are used to construct our pre-training dataset. The corresponding number of 3D volumes for chest, abdomen and head/neck are 2,018, 1,520 and 1,223 respectively. The collection and source details are presented in the supplementary materials. Existing annotations or labels are *not* utilized from these datasets during the pre-training stage.

BTCV : The Beyond the Cranial Vault (BTCV) abdomen challenge dataset [26] contains 30 subjects with abdominal

³<https://www.synapse.org/#!/Synapse:syn3193805/wiki/217785/>

CT scans where 13 organs are annotated by interpreters under supervision of radiologists at Vanderbilt University Medical Center. Each CT scan is acquired with contrast enhancement phase at portal venous consists of 80 to 225 slices with 512×512 pixels and slice thickness ranging from 1 to 6 mm . The multi-organ segmentation problem is formulated as a 13 classes segmentation task (see Table 1 for details). The pre-processing pipeline is detailed in supplementary materials.

MSD: Medical Segmentation Decathlon (MSD) dataset [1] comprises of 10 segmentation tasks from different organs and image modalities. These tasks are designed to feature difficulties across medical images, such as small training sets, unbalanced classes, multi-modality data and small objects. Therefore, the MSD challenge can serve as a comprehensive benchmark to evaluate the generalizability of medical image segmentation methods. The pre-processing pipeline for this dataset is outlined in supplementary materials.

5.2. Implementation Details

For pre-training tasks, (1) masked volume inpainting: the ROI dropping rate is set to 30% (as also used in [2]); the dropped regions are randomly generated and they sum up to reach overall number of voxels; (2) 3D contrastive coding: a feature size of 512 is used as the embedding size; (3) rotation prediction: the rotation degree is configured to 0° , 90° , 180° , and 270° . We train the model using the AdamW [32] optimizer with a warm-up cosine scheduler of 500 iterations. The pre-training experiments use a batch-size of 4 per GPU (with $96 \times 96 \times 96$ patch), and initial learning rate of $4e^{-4}$, momentum of 0.9 and decay of $1e^{-5}$ for 450K iterations. Our model is implemented in PyTorch and MONAI⁴. A five-fold cross validation strategy is used to train models for all BTCV and MSD experiments. We select the best model in each fold and ensemble their outputs for final segmentation predictions. Detailed training hyperparameters for fine-tuning BTCV and MSD tasks can be found in the supplementary materials. All models are trained on a NVIDIA DGX-1 server.

5.3. Evaluation Metrics

The Dice similarity coefficient (Dice) and Hausdorff Distance 95% (HD95) are used as measurements for experiment results. HD95 calculates 95th percentile of surface distances between ground truth and prediction point sets. Metric formulations are as follows:

$$\text{Dice} = \frac{2 \sum_{i=1}^I Y_i \hat{Y}_i}{\sum_{i=1}^I Y_i + \sum_{i=1}^I \hat{Y}_i}, \quad (7)$$

$$\text{HD} = \max \{ \max_{y' \in Y'} \min_{\bar{y}' \in \bar{Y}'} \|y' - \bar{y}'\|, \max_{\bar{y}' \in \bar{Y}'} \min_{y' \in Y'} \|\bar{y}' - y'\| \}. \quad (8)$$

where Y and $\bar{Y}$ denote the ground truth and prediction of voxel values. Y' and $\bar{Y}'$ denote ground truth and prediction

⁴<https://monai.io/>

3D旋转和裁剪同时也可作为对比学习的增强变换。

4.3. 对比编码

自监督对比编码在迁移至下游任务时，于视觉表征学习方面展现出良好性能 [10, 36]。给定一批增强子体积，对比编码通过最大化正样本对（来自同一子体积的增强样本）之间的互信息，同时最小化负样本对（来自不同子体积的视图）之间的互信息，从而实现更好的表征学习。对比编码的实现方式是在Swin UNETR编码器上附加一个线性层，将每个增强子体积映射为潜在表征 v 。如 [10]中所定义，我们使用余弦相似度作为编码表征的距离度量。形式化地，一对 v_i 与 v_j 之间的3D对比编码损失定义为：

$$\mathcal{L}_{contrast} = -\log \frac{\exp(\text{sim}(v_i, v_j)/t)}{\sum_k^{2N} 1_{k \neq i} \exp(\text{sim}(v_i, v_k)/t)}, \quad (5)$$

其中 t 是归一化温度系数的度量。1是当且仅当 $k \neq i$ 时取值为1的指示函数。 sim 表示归一化嵌入之间的点积。对比学习损失函数增强了类内紧致性以及类间可分性。

4.4. 损失函数

形式上，我们通过使用掩码体积修复、3D图像旋转和对比编码等多个预训练目标来训练Swin UNETR的编码器，从而最小化总损失函数，具体如下：

$$\mathcal{L}_{tot} = \lambda_1 \mathcal{L}_{inpaint} + \lambda_2 \mathcal{L}_{contrast} + \lambda_3 \mathcal{L}_{rot}. \quad (6)$$

我们执行了网格搜索超参数优化，以估计 $\lambda_1 = \lambda_2 = \lambda_3 = 1$ 的最优值。

5. 实验

5.1. 数据集

预训练数据集：我们使用共5个公开CT数据集（包含5,050名受试者）来构建预训练数据集。胸部、腹部和头颈部的3D体积对应数量分别为2,018、1,520和1,223。采集和来源详情见补充材料。在预训练阶段，这些数据集中的现有标注或标签未被使用。

BTCV: Beyond the Cranial Vault (BTCV) 腹部挑战数据集 [26] 包含30名受试者，其腹部

³<https://www.synapse.org/#!/Synapse:syn3193805/wiki/217785/>

CT扫描由范德比尔特大学医学中心的放射科医生监督下的标注人员进行13个器官的标注。每次CT扫描均在门静脉对比增强期采集，包含80 至225 个切片，像素为 512×512 ，切片厚度范围为1 至 6 mm 。多器官分割问题被定义为13类别分割任务（详见表格1）。预处理流程详见补充材料。

MSD: 医学分割十项全能（MSD）数据集 [1]包含来自不同器官和图像模态的10项分割任务。这些任务旨在体现医学图像中的各种难点，如训练集小、类别不平衡、多模态数据和小目标等。因此，MSD挑战赛可以作为评估医学图像分割方法泛化能力的综合基准。该数据集的预处理流程详见补充材料。

5.2. 实现细节

对于预训练任务，(1) 掩码体积修复：ROI丢弃率设置为30%（与 [2]中相同）；被丢弃的区域随机生成，其总和达到体素总数；(2) 3D对比编码：使用512的特征大小作为嵌入大小；(3) 旋转预测：旋转角度配置为 0° , 90° , 180° , 和 270° 。我们使用AdamW [32]优化器训练模型，采用500次迭代的预热余弦调度器。预训练实验使用每GPU批大小为4（使用 $96 \times 96 \times 96$ 块），初始学习率为 $4e^{-4}$ ，动量为0.9，衰减为 $1e^{-5}$ ，训练450K次迭代。我们的模型在PyTorch和MONAI⁴中实现。采用五折交叉验证策略为所有BTCV和MSD实验训练模型。我们选择每一折中的最佳模型，并集成其输出以获得最终分割预测。微调BTCV和MSD任务的详细训练超参数可在补充材料中找到。所有模型均在NVIDIA DGX-1服务器上训练。

5.3. 评估指标

Dice相似系数（Dice）和95%豪斯多夫距离（HD95）被用作实验结果的测量指标。HD95计算真实标签与预测点集之间表面距离的 95th 百分位数。指标公式如下：

$$\text{Dice} = \frac{2 \sum_{i=1}^I Y_i \hat{Y}_i}{\sum_{i=1}^I Y_i + \sum_{i=1}^I \hat{Y}_i}, \quad (7)$$

$$\text{HD} = \max \{ \max_{y' \in Y'} \min_{\bar{y}' \in \bar{Y}'} \|y' - \bar{y}'\|, \max_{\bar{y}' \in \bar{Y}'} \min_{y' \in Y'} \|\bar{y}' - y'\| \}. \quad (8)$$

其中 Y 和 $\bar{Y}$ 分别表示体素值的真实标签和预测。 Y' 和 $\bar{Y}'$ 分别表示真实标签和预测

⁴<https://monai.io/>

Methods	Spl	RKid	LKid	Gall	Eso	Liv	Sto	Aor	IVC	Veins	Pan	AG	Avg.
SETR NUP [53]	0.931	0.890	0.897	0.652	0.760	0.952	0.809	0.867	0.745	0.717	0.719	0.620	0.796
SETR PUP [53]	0.929	0.893	0.892	0.649	0.764	0.954	0.822	0.869	0.742	0.715	0.714	0.618	0.797
SETR MLA [54]	0.930	0.889	0.894	0.650	0.762	0.953	0.819	0.872	0.739	0.720	0.716	0.614	0.796
ASPP [8]	0.935	0.892	0.914	0.689	0.760	0.953	0.812	0.918	0.807	0.695	0.720	0.629	0.811
TransUNet [6]	0.952	0.927	0.929	0.662	0.757	0.969	0.889	0.920	0.833	0.791	0.775	0.637	0.838
CoTr* [47]	0.943	0.924	0.929	0.687	0.762	0.962	0.894	0.914	0.838	0.796	0.783	0.647	0.841
CoTr [47]	0.958	0.921	0.936	0.700	0.764	0.963	0.854	0.920	0.838	0.787	0.775	0.694	0.844
RandomPatch [44]	0.963	0.912	0.921	0.749	0.760	0.962	0.870	0.889	0.846	0.786	0.762	0.712	0.844
PaNN [56]	0.966	0.927	0.952	0.732	0.791	0.973	0.891	0.914	0.850	0.805	0.802	0.652	0.854
nnUNet [23]	0.967	0.924	0.957	0.814	0.832	0.975	0.925	0.928	0.870	0.832	0.849	0.784	0.888
UNETR [19]	0.972	0.942	0.954	0.825	0.864	0.983	0.945	0.948	0.890	0.858	0.852	0.812	0.891
Swin UNETR	0.976	0.958	0.956	0.893	0.875	0.985	0.953	0.948	0.904	0.899	0.897	0.846	0.918

Table 1. Leaderboard³Dice results of BTCV challenge on multi-organ segmentation. The proposed method achieves state-of-the-art performance in both free and standard competitions. Note: Spl: spleen, RKid: right kidney, LKid: left kidney, Gall: gallbladder, Eso: esophagus, Liv: liver, Sto: stomach, Aor: aorta, IVC: inferior vena cava, Veins: portal and splenic veins, Pan: pancreas, AG: left and right adrenal glands.

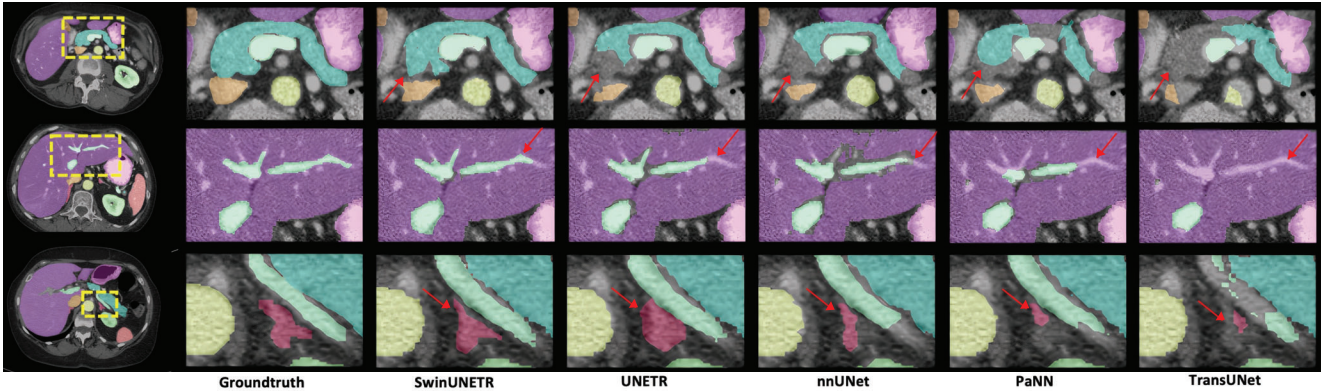


Figure 4. Qualitative visualizations of the proposed Swin UNETR and baseline methods. Three representative subjects are demonstrated. Regions of evident improvements are enlarged to show better details of pancreas (blue), portal vein (light green), and adrenal gland (red).

Method	Rank	Average Accuracy	
		Dice $\uparrow$	NSD $\uparrow$
Swin UNETR	1	78.68	89.28
DiNTS [22]	2	77.93	88.68
nnUNet [23]	3	77.89	88.09
Models Gen. [57]	4	76.97	87.19
Trans VW [18]	5	76.96	87.64

Table 2. Overall performance of top-ranking methods on all 10 segmentation tasks in the MSD public test leaderboard. NSD denotes Normalized Surface Distance.

surface point sets. Surface Dice [33] is also used, which is referred as Normalized Surface Distance (NSD) in MSD challenge evaluation. The metric measures the overlap of ground truth and prediction surfaces (with a fixed tolerance) instead of the overlap of two volumes. This provides a measure of agreement between the surfaces of two structures.

⁵<https://decathlon-10.grand-challenge.org/evaluation/challenge/leaderboard/>

5.4. Results

5.4.1 BTCV Multi-organ Segmentation Challenge

We extensively compare the benchmarks of our model with baselines. The published leaderboard evaluation is shown in Table 1. Compared with other top submissions, the proposed Swin UNETR achieves the best performance. We obtain the state-of-the-art Dice of 0.908, outperforming the second, third and fourth top-ranked baselines by 1.6%, 2.0% and 2.4% on average of 13 organs, respectively. Distinct improvements can be specifically observed for organs that are smaller in size, such as splenic and portal veins of 3.6% against prior state-of-the-art method, pancreas of 1.6%, and adrenal glands of 3.8%. Moderate improvements are observed in other organs. The representative samples in Fig. 4 demonstrate the success of identifying organ details by Swin UNETR. Our method detects the pancreas tail (row 1), and branches in the portal vein (row 2) in Fig. 4, where other methods under segment parts of each tissue. In addition, our method demonstrates distinct improvement in segmentation of adrenal glands (row 3).

方法	Spl	RKid	LKid	Gall	Eso	Liv	Sto	Aor	IVC	静脉	Pan	AG	Avg.
SETR NUP [53]	0.931	0.890	0.897	0.652	0.760	0.952	0.809	0.867	0.745	0.717	0.719	0.620	0.796
SETR PUP [53]	0.929	0.893	0.892	0.649	0.764	0.954	0.822	0.869	0.742	0.715	0.714	0.618	0.797
SETR MLA [54]	0.930	0.889	0.894	0.650	0.762	0.953	0.819	0.872	0.739	0.720	0.716	0.614	0.796
ASPP [8]	0.935	0.892	0.914	0.689	0.760	0.953	0.812	0.918	0.807	0.695	0.720	0.629	0.811
TransUNet [6]	0.952	0.927	0.929	0.662	0.757	0.969	0.889	0.920	0.833	0.791	0.775	0.637	0.838
CoTr* [47]	0.943	0.924	0.929	0.687	0.762	0.962	0.894	0.914	0.838	0.796	0.783	0.647	0.841
CoTr [47]	0.958	0.921	0.936	0.700	0.764	0.963	0.854	0.920	0.838	0.787	0.775	0.694	0.844
RandomPatch [44]	0.963	0.912	0.921	0.749	0.760	0.962	0.870	0.889	0.846	0.786	0.762	0.712	0.844
PaNN [56]	0.966	0.927	0.952	0.732	0.791	0.973	0.891	0.914	0.850	0.805	0.802	0.652	0.854
nnUNet [23]	0.967	0.924	0.957	0.814	0.832	0.975	0.925	0.928	0.870	0.832	0.849	0.784	0.888
UNETR [19]	0.972	0.942	0.954	0.825	0.864	0.983	0.945	0.948	0.890	0.858	0.852	0.812	0.891
Swin UNETR	0.976	0.958	0.956	0.893	0.875	0.985	0.953	0.948	0.904	0.899	0.897	0.846	0.918

表1. 排行榜³BTCV挑战赛在多器官分割上的Dice结果。所提出的方法在自由和标准竞赛中均达到了最先进水平。注：Spl: 脾，RKid: 右肾，LKid: 左肾，Gall: 胆囊，Eso: 食管，Liv: 肝脏，Sto: 胃，Aor: 主动脉，IVC: 下腔静脉，Veins: 门静脉和脾静脉，Pan: 胰腺，AG: 左右肾上腺。

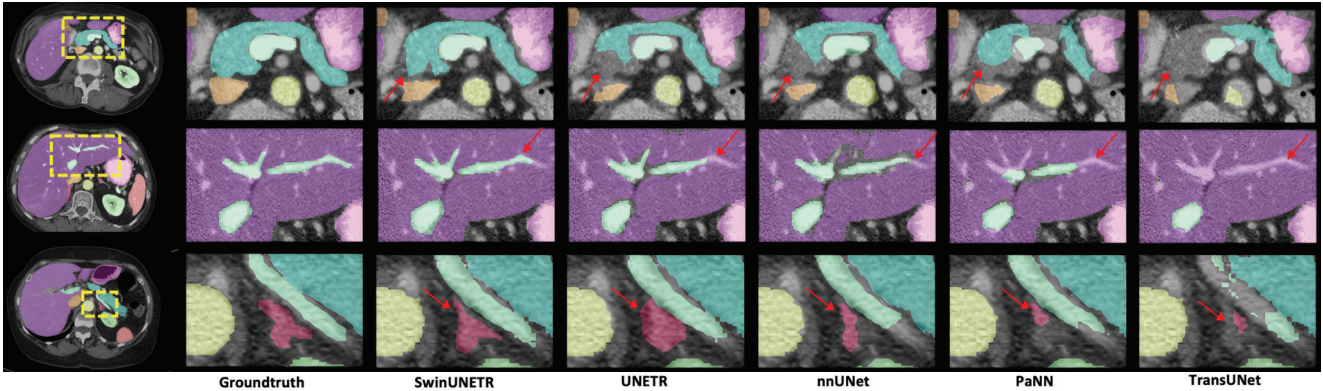


图4. 所提出的Swin UNETR与基线方法的定性可视化。展示了三个代表性受试者。明显改善的区域已放大，以显示胰腺（蓝色）、门静脉（浅绿色）和肾上腺（红色）的更佳细节。

方法	Rank	平均准确率	
		Dice $\uparrow$	NSD $\uparrow$
Swin UNETR	1	78.68	89.28
DiNTS [22]	2	77.93	88.68
nnUNet [23]	3	77.89	88.09
Models Gen. [57]	4	76.97	87.19
Trans VW [18]	5	76.96	87.64

表 2. 在MSD公开测试排行榜上所有10项分割任务中排名靠前方法的整体性能。NSD表示归一化表面距离。

表面点集。还使用了表面Dice [33]，在MSD挑战赛评估中被称为归一化表面距离（NSD）。该指标衡量的是真实标签表面与预测表面之间的重叠（在固定容差下），而非两个体数据之间的重叠。这提供了两种结构表面之间一致性的度量。

⁵<https://decathlon-10.grand-challenge.org/evaluation/challenge/leaderboard/>

5.4. 结果

5.4.1 BTCV多器官分割挑战赛

我们将模型的基准与基线进行了广泛比较。已发布的排行榜评估结果如表1所示。与其他顶级提交结果相比，所提出的 Swin UNETR 取得了最佳性能。我们获得了 0.908 的最先进水平 Dice，在 13 个器官的平均值上，分别比排名第二、第三和第四的顶级基线高出 1.6%、2.0% 和 2.4%。对于体积较小的器官，可以特别观察到显著改进，例如脾静脉和门静脉相比先前最先进方法提高了 3.6%，胰腺提高了 1.6%，肾上腺提高了 3.8%。在其他器官中观察到适度改进。图4中的代表性样本展示了 Swin UNETR 识别器官细节的成功。我们的方法在图4中检测到了胰尾（第 1 行）和门静脉分支（第 2 行），而其他方法则对每种组织的部分区域分割不足。此外，我们的方法在肾上腺分割（第 3 行）方面表现出显著改进。

Organ	Task01 Brain Tumour								Task03 Liver						Task06 Lung	
Metric	Dice1	Dice2	Dice3	Avg.	NSD1	NSD2	NSD3	Avg.	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	NSD
Kim et al [25]	67.40	45.75	68.26	60.47	86.65	72.03	90.28	82.99	94.25	72.96	83.61	96.76	88.58	92.67	63.10	62.51
Trans VW [18]	68.03	46.98	68.40	61.14	87.52	72.42	90.91	83.62	95.18	76.90	86.04	97.86	92.03	94.95	74.54	76.22
C2FNAS [51]	67.62	48.60	69.72	61.98	87.61	72.87	91.16	83.88	94.98	72.89	83.94	98.38	89.15	93.77	70.44	72.22
Models Gen. [57]	68.03	46.98	68.40	61.14	87.52	72.42	90.91	83.62	95.72	77.50	86.61	98.48	91.92	95.20	74.54	76.22
nnUNet [23]	68.04	46.81	68.46	61.10	87.51	72.47	90.78	83.59	95.75	75.97	85.86	98.55	90.65	94.60	73.97	76.02
DiNTS [22]	69.28	48.65	69.75	62.56	89.33	73.16	91.69	84.73	95.35	74.62	84.99	98.69	91.02	94.86	74.75	77.02
Swin UNETR	70.02	52.52	70.51	64.35	89.07	80.30	93.46	87.61	95.35	75.68	85.52	98.34	91.59	94.97	76.60	77.40

Organ	Task07 Pancreas						Task08 Hepatic Vessel						Task09 Spleen		Task10 Colon	
Metric	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	NSD1	Dice1	NSD1
Kim et al [25]	80.61	51.75	66.18	95.83	73.09	84.46	62.34	68.63	65.49	83.22	78.43	80.83	91.92	94.83	49.32	62.21
Trans VW [18]	81.42	51.08	66.25	96.07	70.13	83.10	65.80	71.44	68.62	84.01	80.15	82.08	97.35	99.87	51.47	60.53
C2FNAS [51]	80.76	54.41	67.59	96.16	75.58	85.87	64.30	71.00	67.65	83.78	80.66	82.22	96.28	97.66	58.90	72.56
Models Gen. [57]	81.36	50.36	65.86	96.16	70.02	83.09	65.80	71.44	68.62	84.01	80.15	82.08	97.35	99.87	51.47	60.53
nnUNet [23]	81.64	52.78	67.21	96.14	71.47	83.81	66.46	71.78	69.12	84.43	80.72	82.58	97.43	99.89	58.33	68.43
DiNTS [22]	81.02	55.35	68.19	96.26	75.90	86.08	64.50	71.76	68.13	83.98	81.03	82.51	96.98	99.83	59.21	70.34
Swin UNETR	81.85	58.21	70.71	96.57	79.10	87.84	65.69	72.20	68.95	84.83	81.62	83.23	96.99	99.84	59.45	70.89

Table 3. MSD test dataset performance comparison of Dice and NSD. Benchmarks obtained from MSD test leaderboard⁵.

5.4.2 Segmentation Results on MSD

The overall MSD results per task and ranking from the challenge leaderboard are shown in Table. 2. The proposed Swin UNETR achieves state-of-the-art performance in Task01 BrainTumour, Task06 Lung, Task07 Pancreas, and Task10 Colon. The results are comparable for Task02 Heart, Task03 Liver, Task04 Hippocampus, Task05 Prostate, Task08 HepaticVessel and Task09 Spleen. Overall, Swin UNETR presents the best average Dice of 78.68% across all ten tasks and achieves the top ranking in the MSD leaderboard. The detail number of multiple tasks are shown in Table 3. Qualitative visualization can be observed in Fig. 5. Swin UNETR with self-supervised pre-training demonstrates visually better segmentation results in the CT tasks. The pre-trained weights are only used for fine-tuning CT tasks including Liver, Lung, Pancreas, HepaticVessel, Spleen, and Colon. For MRI tasks: Brain Tumour, Heart, Hippocampus, Prostate, experiments are trained from scratch because of the domain gap between CT and MRI images. Due to space limitations, we present the MSD test benchmarks for the remaining three MRI tasks in the supplementary materials.

5.5. Ablation Study

5.5.1 Efficacy of Pre-training

A comparison of all MSD CT tasks using pre-trained model against training from scratch can be observed in Fig. 6. Distinct improvement can be observed for Task03 Liver, Dice of 77.77% comparing to 75.27%. Task08 Hepatic Vessel achieves 68.52% against 64.63%. Task10 Colon shows the largest improvement, from 34.83% to 43.38%. Task07 Pancreas and Task09 Spleen both achieve significant improvement from 67.12% to 67.82%, and 96.05% to 97.32% respectively.

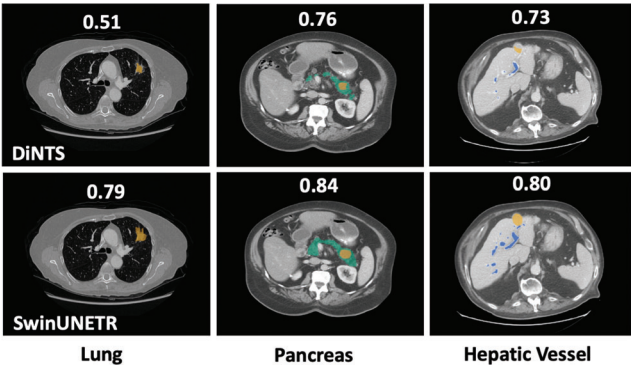


Figure 5. Qualitative results of representative MSD CT tasks. Average Dice values are illustrated on top of each image. Our model demonstrates more accurate performance in comparison to DiNTS for both organ and tumor segmentation across different tasks.

Loss Function	Average Accuracy	
	Dice ↑	HD ↓
Scratch	83.43	42.36
$\mathcal{L}_{\text{rot}}$	83.56	36.19
$\mathcal{L}_{\text{contrast}}$	83.67	38.81
$\mathcal{L}_{\text{inpaint}}$	83.85	28.94
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{rot}}$	84.01	26.06
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{contrast}}$	84.45	24.37
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{contrast}} + \mathcal{L}_{\text{rot}}$	84.72	20.03

Table 4. Ablation study of the effectiveness of each objective function in the proposed pre-training loss. HD denotes Hausdorff Distance. Experiments on fine-tuning the BTCV dataset.

5.5.2 Reduce Manual Labeling Efforts

Fig. 7 demonstrates the comparison results of fine-tuning using a subset of BTCV dataset. We show using 10% of

器官	Task01 脑肿瘤								Task03 肝脏						Task06 肺	
指标	Dice1	Dice2	Dice3	Avg.	NSD1	NSD2	NSD3	Avg.	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	NSD1
Kim 等人 [25]	67.40	45.75	68.26	60.47	86.65	72.03	90.28	82.99	94.25	72.96	83.61	96.76	88.58	92.67	63.10	62.51
Trans VW [18]	68.03	46.98	68.40	61.14	87.52	72.42	90.91	83.62	95.18	76.90	86.04	97.86	92.03	94.95	74.54	76.22
C2FNAS [51]	67.62	48.60	69.72	61.98	87.61	72.87	91.16	83.88	94.98	72.89	83.94	98.38	89.15	93.77	70.44	72.22
Models Gen. [57]	68.03	46.98	68.40	61.14	87.52	72.42	90.91	83.62	95.72	77.50	86.61	98.48	91.92	95.20	74.54	76.22
nnUNet [23]	68.04	46.81	68.46	61.10	87.51	72.47	90.78	83.59	95.75	75.97	85.86	98.55	90.65	94.60	73.97	76.02
DiNTS [22]	69.28	48.65	69.75	62.56	89.33	73.16	91.69	84.73	95.35	74.62	84.99	98.69	91.02	94.86	74.75	77.02
Swin UNETR	70.02	52.52	70.51	64.35	89.07	80.30	93.46	87.61	95.35	75.68	85.52	98.34	91.59	94.97	76.60	77.40

器官	Task07 胰腺						Task08 肝血管						Task09 脾脏		Task10 结肠	
指标	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	Dice2	Avg.	NSD1	NSD2	Avg.	Dice1	NSD1	Dice1	NSD1
Kim 等人 [25]	80.61	51.75	66.18	95.83	73.09	84.46	62.34	68.63	65.49	83.22	78.43	80.83	91.92	94.83	49.32	62.21
Trans VW [18]	81.42	51.08	66.25	96.07	70.13	83.10	65.80	71.44	68.62	84.01	80.15	82.08	97.35	99.87	51.47	60.53
C2FNAS [51]	80.76	54.41	67.59	96.16	75.58	85.87	64.30	71.00	67.65	83.78	80.66	82.22	96.28	97.66	58.90	72.56
Models Gen. [57]	81.36	50.36	65.86	96.16	70.02	83.09	65.80	71.44	68.62	84.01	80.15	82.08	97.35	99.87	51.47	60.53
nnUNet [23]	81.64	52.78	67.21	96.14	71.47	83.81	66.46	71.78	69.12	84.43	80.72	82.58	97.43	99.89	58.33	68.43
DiNTS [22]	81.02	55.35	68.19	96.26	75.90	86.08	64.50	71.76	68.13	83.98	81.03	82.51	96.98	99.83	59.21	70.34
Swin UNETR	81.85	58.21	70.71	96.57	79.10	87.84	65.69	72.20	68.95	84.83	81.62	83.23	96.99	99.84	59.45	70.89

表 3. MSD 测试数据集上 Dice 和 NSD 的性能比较。基准数据来自 MSD 测试排行榜⁵.

5.4.2 MSD 上的分割结果

各任务的总体 MSD 结果以及挑战赛排行榜排名如表 2所示。所提出的 Swin UNETR 在 Task01 脑肿瘤、Task06 肺、Task07 胰腺和 Task10 结肠中取得了最先进水平的性能。对于 Task02 心脏、Task03 肝脏、Task04 海马体、Task05 前列腺、Task08 肝血管和 Task09 脾脏，结果相当。总体而言，Swin UNETR 在所有十项任务中呈现出 78.68% 的最佳平均 Dice，并在 MSD 排行榜中取得最高排名。多个任务的详细数据如表3所示。定性可视化可在图5中观察到。经过自监督预训练的 Swin UNETR 在 CT 任务中展现出视觉上更好的分割结果。预训练权重仅用于微调 CT 任务，包括肝脏、肺、胰腺、肝血管、脾和结肠。对于 MRI 任务：脑肿瘤、心脏、海马体、前列腺，由于 CT 和 MRI 图像之间存在域差距，实验采用从头训练。由于篇幅限制，我们在补充材料中展示了其余三项 MRI 任务的 MSD 测试基准。

5.5. 消融研究

5.5.1 预训练的有效性

使用预训练模型与从头训练在所有 MSD CT 任务上的比较见图6。Task03 肝脏取得了显著提升，Dice 从 75.27% 提升至 77.77%。Task08 肝血管从 64.63% 提升至 68.52%。Task10 结肠的提升最为显著，从 34.83% 提升至 43.38%。Task07 胰腺和 Task09 脾脏也均取得了显著提升，分别从 67.12% 提升至 67.82%，以及从 96.05% 提升至 97.32%。

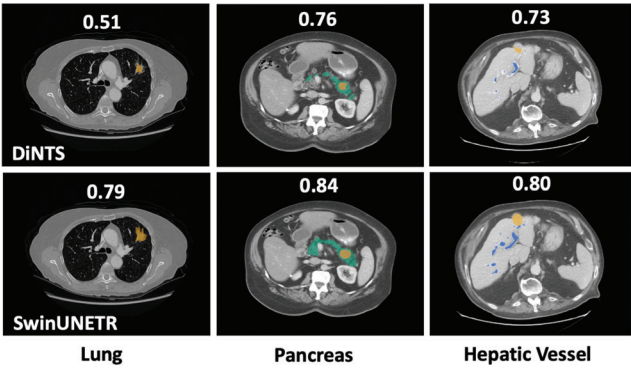


图 5. 代表性 MSD CT 任务的定性结果。平均 Dice 值标注在每幅图像上方。与 DiNTS 相比，我们的模型在不同任务中的器官分割和肿瘤分割方面均表现出更准确的性能。

损失函数	平均准确率	
	Dice ↑	HD ↓
从头训练	83.43	42.36
$\mathcal{L}_{\text{rot}}$	83.56	36.19
$\mathcal{L}_{\text{对比}}$	83.67	38.81
$\mathcal{L}_{\text{inpaint}}$	83.85	28.94
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{rot}}$	84.01	26.06
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{contrast}}$	84.45	24.37
$\mathcal{L}_{\text{inpaint}} + \mathcal{L}_{\text{contrast}} + \mathcal{L}_{\text{rot}}$	84.72	20.03

表 4. 所提出的预训练损失中每个目标函数有效性的消融研究。HD表示豪斯多夫距离。在BTCV数据集上进行微调的实验。

5.5.2 减少手动标注工作量

图 7 展示了使用BTCV数据集子集进行微调的比较结果。我们展示了使用10%的

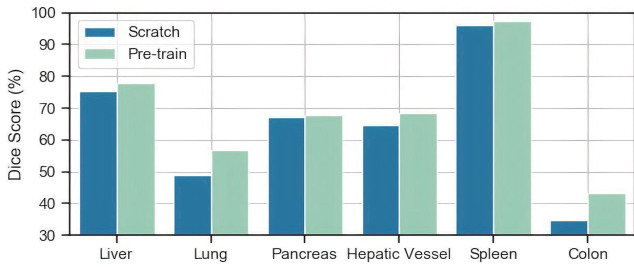


Figure 6. The indication of Dice gap between using pre-training (Green) and scratch model (Blue) on MSD CT tasks validation set.

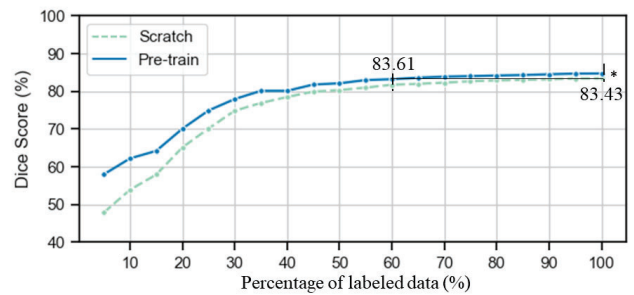


Figure 7. Data-efficient performance on BTCV test dataset. Significance under Wilcoxon Signed Rank test, *: $p < 0.001$.

labeled data, experiments with pre-training weights achieve approximately 10% improvement comparing to training from scratch. On employing all labeled data, the self-supervised pre-training shows 1.3% higher average Dice. The Dice number 83.13 of learning from scratch with entire dataset can be achieved by using pre-trained Swin UNETR with 60% data. Fig. 7 indicates that our approach can reduce the annotation effort by at least 40% for BTCV task.

5.5.3 Size of Pre-training Dataset

We perform organ-wise study on BTCV dataset by using pre-trained weights of smaller unlabeled data. In Fig. 8, the fine-tuning results are obtained from pre-training 100, 3,000, and 5,000 scans. We observe that Swin UNETR is robust with respect to the total number of CT scans trained. Fig. 8 demonstrates the proposed model can benefit from larger pre-training datasets with increasing size of unlabeled data.

5.5.4 Efficacy of Self-Supervised Objectives

We perform empirical study on pre-training with different combinations of self-supervised objectives. As shown in Table 4, on BTCV test set, using pre-trained weights by inpainting achieves the highest improvement at single task modeling. On pairing tasks, inpainting and contrastive learning show Dice of 84.45% and Hausdorff Distance (HD) of 24.37. Overall, employing all proxy tasks achieves best Dice of 84.72%.

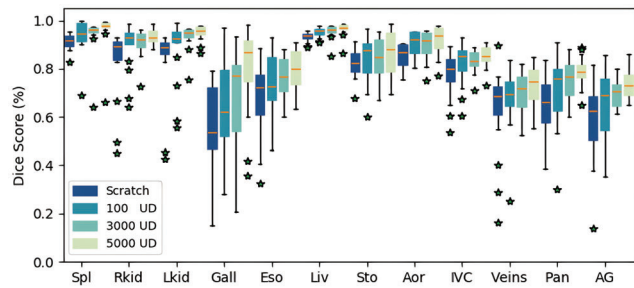


Figure 8. Pre-trained weights using 100, 3000 and 5000 scans are compared for fine-tuning on the BTCV dataset for each organ.

6. Discussion and Limitations

Our state-of-the-art results on the test leaderboards of MSD and BTCV datasets validate the effectiveness of the proposed self-supervised learning framework in taking the advantage of large number of available medical images without the need of annotation effort. Subsequently, fine-tuning the pretrained Swin UNETR model achieves higher accuracy, improves the convergence speed, and reduces the annotation effort in comparison to training with randomly initialized weights from scratch. Our framework is scalable and can be easily extended with more proxy tasks and augmentation transformations. Meanwhile, the pre-trained encoder can benefit the transfer learning of various medical imaging analysis tasks, such as classification and detection. In MSD pancreas segmentation task, Swin UNETR with pre-trained weights outperforms AutoML algorithms such as DiNTS [22] and C2FNAS [51] that are specifically designed for searching the optimal network architectures on the same segmentation task. Currently, Swin UNETR has only been pre-trained using CT images, and our experiments have not demonstrated enough transferability when applied directly to other medical imaging modalities such as MRI. This is mainly due to obvious domain gaps and different number of input channels that are specific to each modality. As a result, this is a potential direction that should be studied in future efforts.

7. Conclusions

In this work, we present a novel framework for self-supervised pre-training of 3D medical images. Inspired by merging feature maps at scales, we built the Swin UNETR by exploiting transformer-encoded spatial representations into convolution-based decoders. By proposing the first transformer-based 3D medical image pre-training, we leverage the power of Swin Transformer encoder for fine-tuning segmentation tasks. Swin UNETR with self-supervised pre-training achieves the state-of-the-art performance on the BTCV multi-organ segmentation challenge and MSD challenge. Particularly, we presented the large-scale CT pre-training with 5,050 volumes, by combining multiple publicly available datasets and diversities of anatomical ROIs.

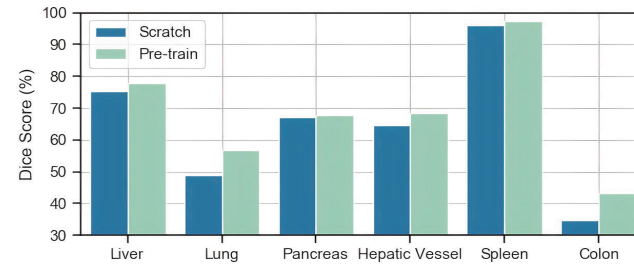


图 6. 在 MSD CT 任务验证集上，使用预训练（绿色）与从头训练模型（蓝色）之间的 Dice 差距指示。

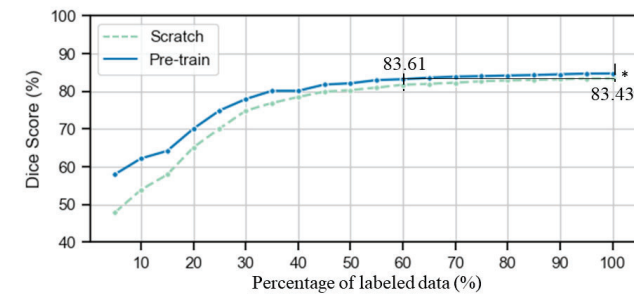


图 7. 在 BTCV 测试数据集上的数据高效性能。Wilcoxon 符号秩检验下的显著性，*: $p < 0.001$ 。

标注数据时，使用预训练权重的实验相比从头训练取得了约10%的提升。在使用全部标注数据时，自监督预训练显示出1.3%更高的平均Dice。使用全部数据集从头训练所达到的Dice值83.13，可以通过使用预训练的Swin UNETR和60%的数据来实现。图 7 表明我们的方法可以将BTCV任务的标注工作量至少减少40%。

5.5.3 预训练数据集规模

我们通过使用较小无标注数据的预训练权重，在 BTCV 数据集上进行逐器官研究。在图 8 中，微调结果分别来自对 100、3,000 和 5,000 次扫描的预训练。我们观察到，Swin UNETR 对于所训练的 CT 扫描总数具有鲁棒性。图 8 表明，随着无标注数据规模的增加，所提出的模型能够从更大的预训练数据集中受益。

5.5.4 自监督目标的有效性

我们对不同自监督目标组合的预训练进行了实证研究。如表4所示，在BTCV测试集上，使用图像修复的预训练权重在单任务建模中取得了最高提升。在配对任务中，图像修复和对比学习的Dice为84.45%，豪斯多夫距离（HD）为24.37。总体而言，采用所有代理任务取得了最佳Dice，为84.72%。

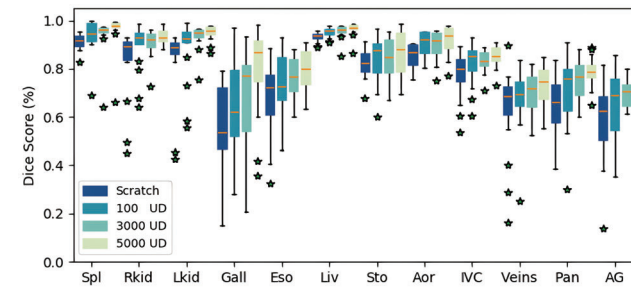


图 8. 使用 100、3000 和 5000 次扫描的预训练权重在 BTCV 数据集上针对每个器官进行微调的对比。

6. 讨论与局限性

我们在MSD和BTCV数据集测试排行榜上取得的最先进水平结果，验证了所提出的自监督学习框架在利用大量可用医学图像而无需标注工作方面的有效性。随后，微调预训练的Swin UNETR模型相比使用随机初始化权重从头训练，能够获得更高的准确率、加快收敛速度并减少标注工作。我们的框架具有可扩展性，可以轻松扩展更多代理任务和增强变换。同时，预训练编码器可以有利于各种医学影像分析任务的迁移学习，例如分类和检测。在MSD胰腺分割任务中，使用预训练权重的Swin UNETR优于AutoML算法，如 DiNTS [22] 和 C2FNAS [51]，这些算法专门为在同一分割任务上搜索最优网络架构而设计。目前，Swin UNETR仅使用CT图像进行了预训练，我们的实验尚未证明在直接应用于其他医学成像模式（如MRI）时具有足够的可迁移性。这主要是由于明显的域差距以及每种模式特定的不同输入通道数所致。因此，这是未来工作中应研究的一个潜在方向。

7. 结论

在这项工作中，我们提出了一种用于3D医学图像自监督预训练的新框架。受多尺度特征图融合的启发，我们通过将Transformer编码的空间表示融入基于卷积的解码器，构建了Swin UNETR。通过提出首个基于Transformer的3D医学图像预训练，我们利用 Swin Transformer编码器的能力来微调分割任务。经过自监督预训练的Swin UNETR在BTCV多器官分割挑战赛和MSD挑战赛上取得了最先进的性能。特别地，我们通过结合多个公开数据集和多种解剖感兴趣区，提出了包含5,050个体数据的大规模CT预训练。

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